Nose — Sinusitis, DNS & Epistaxis

ENT | 3rd Year MBBS

Hi Sammy!
Nose = easy marks!
DNS + Epistaxis are
asked every year!

Nose | DNS | Sinusitis | Epistaxis | ENT

What's Inside

- Deviated Nasal Septum — Types & Management p.2
- Sinusitis — Acute, Chronic & Complications p.3
- Epistaxis — Causes & Management p.4
- Clinical Tests & Procedures p.5
- Exam Strategy & High Yield p.6

Deviated Nasal Septum (DNS)

DNS = most common cause of nasal obstruction! Learn types!

DEFINITION

Deviation of nasal septum from midline causing nasal obstruction.

Types of Deviated Nasal Septum

Type	Description
C-shaped	Simple deviation to one side (most common type)
S-shaped	Deviation to both sides — obstructs both nasal cavities
Spur	Sharp bony/cartilaginous projection — causes epistaxis & headache
Thickening	Localized thickening of septal cartilage or bone
Subluxation	Lower edge of septum displaced from maxillary crest

Clinical Features of DNS

Nasal obstruction	Most common symptom — unilateral or alternating
Headache	Spur contact with turbinate (Sluder's neuralgia)
Epistaxis	Spur stretches mucosa, crusting, bleeding on removal
Sinusitis	Blocked sinus ostia -> recurrent infections
External deformity	Deviated external nose in severe cases
Anosmia	Air cannot reach olfactory area in severe deviation

SMR vs Septoplasty (The Comparison)

Feature	SMR (Killian's)	Septoplasty
Approach	Removes deviated cartilage/bone	Reshapes & repositions septum
Preservation	Less (more cartilage removed)	More (cartilage preserved)
Complications	Septal perforation, saddle nose	Less risk of complications
Current status	Largely replaced	Procedure of choice today
Age	After 17 years (growth)	After 17 years (growth)

Cottle's Test!
Pull cheek laterally -> if breathing improves, test is POSITIVE -> obstruction at internal nasal valve. Cottle classified septum into 5 areas!

Sinusitis complications
= life-threatening!
Don't miss this!

Acute vs Chronic Sinusitis

Feature	Acute Sinusitis	Chronic Sinusitis
Duration	< 12 weeks	> 12 weeks
Etiology	Viral -> bacterial (S. pneumo)	Allergy, polyps, fungi
Symptoms	Facial pain, purulent discharge	Post-nasal drip, nasal block
XR finding	Air-fluid level, opacity	Mucosal thickening
Treatment	Antibiotics + decongestants	FESS if medical Rx fails

Maxillary Sinusitis (Most Common)

- Maxillary sinus = largest paranasal sinus, first to develop
- Natural ostium opens in middle meatus (hiatus semilunaris)
- Dental infections (premolars/molars) can cause odontogenic sinusitis
- Proof puncture: through inferior meatus for irrigation (antral wash)

Complications of Sinusitis (HIGH YIELD!)

ORBITAL (Chandler's)

- Preseptal cellulitis (most common)
- Orbital cellulitis (post-septal)
- Subperiosteal abscess
- Orbital abscess
- Cavernous sinus thrombosis

INTRACRANIAL

- Meningitis (most common IC)
- Epidural abscess
- Subdural empyema
- Brain abscess (frontal lobe)
- Cavernous sinus thrombosis

FESS (Functional Endoscopic Sinus Surgery)

1. Gold standard for chronic sinusitis refractory to medical treatment
2. Principle: Restore ventilation & drainage of sinuses via natural ostia
3. Key landmark: Uncinate process -> removed first (uncinectomy)

★ **Ostiomeatal Complex (OMC)**
OMC = final common pathway for drainage of maxillary, anterior ethmoid & frontal sinuses.

Little's area vs
Woodruff's plexus —
know the difference!

Little's Area (Kiesselbach's Plexus)

- Location: Anteroinferior part of nasal septum
- 4 arteries anastomose: Anterior ethmoidal, Sphenopalatine, Superior labial, Greater palatine arteries
- Most common site of epistaxis (90% of nosebleeds)

Causes of Epistaxis

Category	Causes
Local	Trauma (nose picking #1), DNS spur, rhinitis, tumors (JNA, NPC)
Systemic	Hypertension (#1 systemic), liver disease, anticoagulants
Blood disorders	Thrombocytopenia, hemophilia, von Willebrand, leukemia
Hereditary	Osler-Weber-Rendu (HHT) — telangiectasias + epistaxis
Idiopathic	Most common overall cause in clinical practice

Management of Epistaxis (Stepwise)

1	First Aid	Pinch nose (Little's area), lean forward, ice on bridge
2	Cautery	Silver nitrate / electrocautery for visible bleeding point
3	Anterior Pack	Ribbon gauze soaked in liquid paraffin or Merocel
4	Posterior Pack	Foley catheter / Bellocq's tampon for posterior bleed
5	Surgical	Arterial ligation (sphenopalatine, anterior ethmoidal, ECA)
6	Embolization	Angiographic embolization for refractory epistaxis

Anterior vs Posterior Epistaxis

Feature	Anterior	Posterior
Site	Little's area (Kiesselbach)	Woodruff's plexus
Age	Young patients	Elderly / hypertensives
Frequency	90% of all nosebleeds	10% of all nosebleeds
Severity	Usually mild, self-limiting	Severe, may need surgery
Artery	Anterior ethmoidal branches	Sphenopalatine artery
Treatment	Cautery / anterior packing	Posterior packing / ligation

These tests come in
viva every single time!
Know them well!

Important Clinical Tests — Nose

Test	Description & Significance
Cottle's test	Pull cheek laterally — positive if breathing improves (nasal valve)
Cold spatula test	Misting on metal spatula — tests nasal patency each nostril
Proof puncture	Antral puncture via inferior meatus — diagnostic + therapeutic
Water's view XR	Occipito-mental view — best plain XR for maxillary sinuses
CT PNS	Gold standard imaging for sinusitis, pre-FESS evaluation
Displacement test	Proetz's method — sinus irrigation by negative pressure

Nasal Polyps — Quick Review

- Ethmoidal polyps = most common, bilateral, allergic, insensitive to touch
- Antrochoanal polyp = unilateral, from maxillary sinus, young males
- Samter's triad: Nasal polyps + Aspirin sensitivity + Asthma
- Treatment: Topical steroids (fluticasone); FESS if recurrent/severe

Rhinitis — Types at a Glance

Type	Key Feature	Treatment
Allergic	Sneezing, watery rhinorrhea, itch	Antihistamines, nasal steroids
Vasomotor	Non-allergic, autonomic dysfunction	Ipratropium bromide spray
Atrophic	Foul smell (ozaena), crusting	Nasal douche, glucose-glycerin drops
Rhinitis sicca	Dry anterior septum, crusting	Saline drops, petroleum jelly
Medicamentosa	Rebound from decongestant overuse	Stop decongestant, nasal steroids

★ Atrophic Rhinitis!
Young female + foul smell (ozaena) + spacious nasal cavity + greenish crusts
= ATROPHIC RHINITIS. Klebsiella ozaenae is the organism. Treatment: Young's operation.

Nose = guaranteed
15+ marks in ENT!
You've GOT this Sammy!

Most Repeated LAQs (10–12 marks)

- 1. Classify DNS. Describe clinical features & management
- 2. Describe acute & chronic sinusitis with complications
- 3. Enumerate causes of epistaxis. Describe management (stepwise)
- 4. Write about FESS — indications, principle & complications
- 5. Describe nasal polyps — types, features & management

Most Repeated SAQs (5–6 marks)

- 1. Cottle's test — technique & significance
- 2. Little's area — anatomy & clinical importance
- 3. Anterior vs posterior nasal packing
- 4. Complications of sinusitis (Chandler's classification)
- 5. Samter's triad
- 6. Atrophic rhinitis — features & management

One-Liner Associations (MCQ/Viva Gold)

Question	Answer
Most common cause of nasal obstruction	Deviated nasal septum (DNS)
Most common site of epistaxis	Little's area (Kiesselbach's plexus)
Most common organism in acute sinusitis	Streptococcus pneumoniae
Most common paranasal sinus involved	Maxillary sinus
Antral puncture is done through	Inferior meatus
Atrophic rhinitis organism	Klebsiella ozaenae
Samter's triad	Polyps + aspirin sensitivity + asthma
Unilateral nasal polyp in young male	Antrochoanal polyp
Unilateral bloody nasal discharge	Think malignancy / JNA
Osler-Weber-Rendu disease	Hereditary hemorrhagic telangiectasia
Proof puncture is done via	Inferior meatus
Best imaging for sinusitis	CT scan (coronal cuts)

♥ Every patient who breathes better thanks you, Sammy!
The nose is the gateway to the airway. Master it well!